

High School Students

HIGH SCHOOL STUDENT MODULE (CLASS 9–12)

PROJECT: TALENTSPHERE ELEVATE

PURPOSE:

The High School Student Module is designed to help students discover suitable career paths, understand their strengths and interests, and build future-ready skills through AI-powered recommendations and personalized learning roadmaps.

1. STUDENT REGISTRATION & PROFILE

Students create an account and provide:

- Name
- Class (9–12)
- School Name
- Age
- Preferred Language
- Favorite Subjects
- Hobbies
- Career Interests

Example:

Name: Arun Kumar

Class: 10

Favorite Subjects: Mathematics, Science

Hobbies: Robotics, Gaming

Career Interest: Technology

2. CAREER INTEREST ASSESSMENT

The system asks questions to understand the student's interests and preferences.

Sample Questions:

1. Which activity do you enjoy most?
 - Solving puzzles

- Drawing designs
 - Helping people
 - Building things
2. What would you like to create?
- Mobile App
 - Machine
 - Artwork
 - Business
3. Which subject excites you the most?
- Mathematics
 - Physics
 - Biology
 - Computer Science

AI analyzes responses and calculates career domain scores.

Example:

Technology: 92%

Engineering: 88%

Design: 45%

Healthcare: 38%

3. PERSONALITY & APTITUDE ANALYSIS

The platform evaluates:

- Logical Thinking
- Creativity
- Communication Skills
- Problem Solving
- Teamwork
- Leadership Potential

Example Result:

Logical Thinking: 95%

Problem Solving: 91%

Creativity: 72%

Communication: 68%

Leadership: 70%

Best Career Match:
AI & Software Engineering

4. AI CAREER SUGGESTIONS

Based on interests, aptitude, and academic strengths, the system recommends suitable careers.

Example Recommendations:

AI Engineer - 95% Match
Software Developer - 92% Match
Robotics Engineer - 88% Match
Data Scientist - 84% Match

Reason for Recommendation:

- **Strong mathematics performance**
 - **High logical reasoning ability**
 - **Interest in technology**
 - **Excellent problem-solving skills**
-

5. FUTURE SKILLS ROADMAP

The AI creates a personalized learning path from school level to college level.

Example: AI Engineer Career Path

Class 10:

- **Python Basics**
- **Mathematics Foundation**
- **Logical Reasoning**
- **Scratch Programming**

Class 11:

- **Intermediate Python**
- **Statistics**
- **Data Handling**

- **Mini Projects**

Class 12:

- **Data Structures**
 - **Machine Learning Basics**
 - **AI Concepts**
 - **Portfolio Projects**
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6. LEARNING RECOMMENDATIONS

The platform recommends courses, videos, projects, and practice exercises.

Examples:

Python → Python for Beginners

Logic Building → 100 Logic Questions

AI Basics → Introduction to Artificial Intelligence

Mathematics → Algebra & Statistics Practice

7. GOAL TRACKING DASHBOARD

Students can set learning goals and monitor progress.

Example Goals:

Complete Python Basics - 80%

Finish 3 Mini Projects - 66%

Learn AI Fundamentals - 40%

Additional Metrics:

Projects Completed: 2

Learning Hours This Week: 12

Badges Earned: 5

8. SKILL DEVELOPMENT TRACKER

The platform continuously tracks skill growth.

Example Skill Levels:

Python: 75%

Mathematics: 82%

Logical Reasoning: 90%

AI Concepts: 25%

Students can view progress charts and improvement trends over time.

9. HIGH SCHOOL DASHBOARD

Dashboard Overview:

Welcome, Arun

Class: 10

Career Match Score: 95%

Learning Progress: 72%

Goals Completed: 4/6

Badges Earned: 5

Top Career Recommendation:

AI Engineer (95% Match)

Why This Career?

- **Strong logical reasoning**
- **Excellent mathematics skills**
- **Interest in technology**
- **Good problem-solving ability**

Next 3-Month Learning Plan:

Month 1:

- **Python Basics**

Month 2:

- **Logic Building**

Month 3:

- **AI Fundamentals**

Skill Progress:

Python: 75%

Mathematics: 82%

Logical Reasoning: 90%

AI Concepts: 25%

EXPECTED OUTCOME

The High School Student Module helps students:

- **Discover suitable careers**
- **Understand their strengths and weaknesses**
- **Learn future industry skills**
- **Follow a personalized learning roadmap**
- **Track goals and progress**
- **Prepare early for higher education and future careers**

This module transforms TalentSphere Elevate from a simple career platform into an AI-powered career guidance and future readiness system for school students.

College Students

COLLEGE STUDENT MODULE (UG / PG)

PROJECT: TALENTSPHERE ELEVATE

PURPOSE

The College Student Module is designed to help undergraduate and postgraduate students prepare for placements, improve coding skills, build ATS-friendly resumes, identify skill gaps, and receive AI-powered internship and job recommendations.

1. STUDENT PROFILE

Students provide:

- Name
- College Name
- Degree (B.E, B.Tech, B.Sc, MCA, etc.)
- Department
- Year of Study
- CGPA / Percentage
- Skills Known
- Interested Job Roles
- Resume Upload

Example:

- Name: Arun Kumar
- Degree: B.Tech
- Department: Computer Science
- Year: 3rd Year
- Skills: Python, HTML, CSS
- Interested Role: Software Developer

2. ATS-FRIENDLY RESUME BUILDER

Features:

- Pre-designed ATS templates

- Auto-formatting
- Education section
- Skills section
- Projects section
- Internship section
- Download as PDF/DOCX

AI Suggestions:

- Add measurable project outcomes
- Include relevant technical keywords
- Improve summary statement
- Remove unnecessary information

3. AI RESUME REVIEW

After uploading a resume, the AI evaluates:

Criteria	Score
ATS Compatibility	85%
Keyword Optimization	78%
Project Quality	82%
Formatting	90%

Overall Resume 84%
Score

AI Feedback Example:

"Your resume is ATS-friendly but lacks SQL and Data Structures keywords required for Software Developer roles."

4. CODING PRACTICE TESTS

Topics:

- Python
- Java
- C/C++
- Data Structures
- Algorithms
- SQL
- Web Development

Test Result Example:

Topic	Score
Python	82%
Data Structures	55%
SQL	48%

Algorithms 60%

5. PLACEMENT READINESS SCORE

The AI calculates a readiness score based on:

- Resume Quality
- Coding Test Performance
- Projects
- Communication Skills
- Skill Match
- Internship Experience

Placement Readiness Formula:

- Resume: 25%
- Coding: 30%
- Projects: 20%
- Communication: 15%
- Experience: 10%

Final Placement Score: 78/100

6. MOCK INTERVIEW SYSTEM

Sample Questions:

- Explain OOP concepts.
- What is a REST API?
- How does SQL JOIN work?
- Describe a project you built.
- How would you optimize a slow program?

AI Evaluates:

- Answer relevance
- Confidence level
- Technical accuracy

- Communication quality

7. SKILL GAP ANALYSIS

Target Role: Software Developer

Required Skill	Status
Python	Available
SQL	Missing
Data Structures	Missing
Git	Basic Level

AI Recommendation:

- Learn SQL fundamentals
- Practice Data Structures daily
- Build one database project
- Use GitHub regularly

8. INTERNSHIP RECOMMENDATIONS-optional

Company	Role
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Infosys Python Intern

TCS Software Trainee

Wipro Application Support
Intern

Zoho Backend Intern

9. JOB MATCHING ENGINE

Matching Factors:

- Skills
- CGPA
- Projects
- Resume Score
- Coding Performance
- Location Preference

Example Job Matches:

Job Role	Match %
Software Developer	88%

Backend Developer	84%
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Data Analyst	76%
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QA Engineer	72%
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10. AI OUTPUT EXAMPLE

Placement Ready Evaluation

Placement Score	78/100
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Resume ATS Score	85%
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Coding Score	72%
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Job Match	88%
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Missing Skills:

- Data Structures
- SQL
- Git Advanced Commands

Strong Areas:

- Python Programming

- Project Development
- Problem Solving

Suggested Companies:

- Infosys
- TCS
- Wipro
- Zoho

30-Day Action Plan:

Week	Focus
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Week 1	Learn SQL Basics
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Week 2	Practice Data Structures
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Week 3	Build CRUD Project
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Week 4	Attend Mock Interview
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11. COLLEGE DASHBOARD

Dashboard Overview:

- Placement Score: 78/100
- Resume ATS Score: 85%
- Coding Score: 72%

- Job Match: 88%

Top Recommendation:

Software Developer (88% Match)

Why This Role?

- Strong Python skills
- Good project portfolio
- High problem-solving ability
- Excellent ATS resume score

Focus This Month:

- SQL Basics
- Data Structures
- CRUD Project Development

Current Skill Progress:

Skill	Progress
Python	82%
Data Structures	55%
SQL	48%
Algorithms	60%

EXPECTED OUTCOME

The College Student Module helps students:

- Build ATS-friendly resumes
- Improve coding performance
- Identify missing skills
- Prepare for placements
- Practice mock interviews
- Find internships
- Match with suitable jobs
- Increase placement readiness

FINAL PROJECT STATEMENT

“The College Student Module of TalentSphere Elevate uses AI to evaluate resumes, analyze coding performance, identify skill gaps, calculate placement readiness, recommend internships, and match students with suitable job opportunities, thereby helping them become industry-ready professionals.”

Working Professionals

Below is the Working Professionals module in simple TXT format so you can easily copy, share, or include it in your project documentation.

WORKING PROFESSIONAL MODULE (EXPERIENCED EMPLOYEES)

PROJECT: TALENTSPHERE ELEVATE

PURPOSE

The Working Professional Module is designed to help experienced employees improve their skills, transition to better career opportunities, prepare for promotions, and receive AI-powered professional growth recommendations.

1. PROFESSIONAL PROFILE

Users provide:

- **Name**
- **Current Company**
- **Current Role**
- **Total Experience**
- **Technical Skills**
- **Leadership Experience**
- **Certifications**
- **Career Goals**
- **Preferred Job Roles**

Example:

- **Name: Arun Kumar**
- **Current Role: Software Developer**
- **Experience: 4 Years**
- **Skills: Python, Django, REST API**
- **Career Goal: Become Senior Backend Engineer**

2. PROFESSIONAL SKILL ASSESSMENT

The AI evaluates:

Skill Area	Score
Backend Development	88%
System Design	72%
Cloud Computing	55%
DevOps	48%
Leadership	70%

3. AI CAREER TRANSITION SUGGESTIONS

Based on experience and skills, the system recommends next-level roles.

Current Role: Software Developer

Suggested Career Paths:

Next Role	Match %
Senior Backend Engineer	92%

Backend Architect 81%

Cloud Engineer 74%

Engineering Lead 69%

4. PROMOTION READINESS ANALYSIS

The AI checks:

- **Years of experience**
- **Project complexity**
- **Leadership exposure**
- **Technical expertise**
- **Communication skills**
- **Team contributions**

Example Result:

Promotion 78
Readiness %

Technical Readiness 85
%

Leadership 70
Readiness %

Communication	76
Readiness	%

5. SALARY BENCHMARK INSIGHTS

Example:

Current Salary	₹6 LPA
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Market Average	₹7.5 LPA
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Senior Backend Engineer Range	₹9–12 LPA
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Potential Growth	+25%
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6. INDUSTRY TREND RECOMMENDATIONS

Trending Skills:

- Cloud Computing
- Docker
- Kubernetes
- Microservices
- AI Integration
- System Design

Industry Demand Forecast:

Skill	Demand Growth
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Kubernetes	+38%
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AWS	+32%
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Microservices	+28%
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System Design	+25%
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7. CERTIFICATION SUGGESTIONS

Recommended Certifications:

Certification	Priority
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AWS Solutions Architect	High
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Docker Certified Associate	High
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Kubernetes Administrator	Medium
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System Design Fundamentals	Medium
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8. LEADERSHIP SKILL EVALUATION

Evaluation Areas:

- **Team Coordination**
- **Mentoring**
- **Decision Making**
- **Conflict Resolution**
- **Project Ownership**

Example Result:

Area	Score
Team Coordination	78%
Mentoring	72%
Decision Making	80%

Project 75%
Ownership

9. ADVANCED JOB MATCHING

Matching Factors:

- **Experience Level**
- **Technical Skills**
- **Leadership Ability**
- **Industry Domain**
- **Certifications**
- **Salary Expectation**

Example Job Matches:

Role	Match %
Senior Backend Engineer	92%
Backend Architect	81%
Cloud Engineer	74%
Engineering Lead	69%

10. AI OUTPUT EXAMPLE

Growth Opportunity Analysis

AI Generated

Current Role

Software Developer

Next Best Role

Senior Backend Engineer

Promotion Ready

78%

Salary Growth

+25%

Strong Areas

Keep Building

- **Backend Development**
- **API Design**
- **Problem Solving**
- **Project Delivery**

Improvement Areas

Focus Next

- **Cloud Computing**
- **Docker**
- **Kubernetes**
- **System Design**

Recommended Certifications

High Priority

AWS Docker Kubernetes

90-Day Action Plan

AI Roadmap

Month	Focus
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Month 1	AWS Fundamentals
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Month 2	Docker & Kubernetes Basics
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Month 3	System Design & Microservices
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11. PROFESSIONAL DASHBOARD

Dashboard Overview:

- **Current Role:** Software Developer
- **Experience:** 4 Years
- **Promotion Readiness:** 78%
- **Salary Growth Potential:** +25%
- **Top Job Match:** Senior Backend Engineer (92%)

Recommended Next Steps:

- **Complete AWS certification**
- **Learn Docker & Kubernetes**
- **Study System Design**
- **Lead one backend project**

Current Skill Progress:

Skill	Progress
Backend Development	88%
System Design	72%
Cloud Computing	55%
DevOps	48%
Leadership	70%

EXPECTED OUTCOME

The Working Professional Module helps users:

- Identify career growth opportunities
- Prepare for promotions
- Transition to higher-paying roles
- Learn industry-demand skills
- Earn relevant certifications
- Improve leadership capabilities
- Match with advanced job opportunities
- Increase salary potential

FINAL PROJECT STATEMENT

“The Working Professional Module of TalentSphere Elevate uses AI to analyze professional skills, identify growth opportunities, recommend career transitions, evaluate promotion readiness, suggest certifications,

provide salary benchmark insights, and match experienced employees with advanced career opportunities.”

PROJECT PROMPT

Build a production-ready full-stack AI-powered platform called “TalentSphere Elevate” with three user roles: High School Students, College Students, and Working Professionals. High School Students should receive AI-based career discovery, future skill roadmaps, and learning recommendations. College Students should receive ATS-friendly resume building, placement readiness scoring, coding assessments, skill gap analysis, internship recommendations, and job matching. Working Professionals should receive skill assessments, career transition recommendations, promotion readiness analysis, certification suggestions, salary growth insights, and advanced job matching. Use FastAPI for the backend, Streamlit or React for the frontend, PostgreSQL for the database, and spaCy/NLTK for AI and NLP processing. Implement secure role-based authentication, resume upload and parsing, personalized AI recommendations, analytics dashboards, notification systems, and exportable reports. The platform should function as a unified Career Growth, Placement, and Recruitment Intelligence System that supports users from school to professional career advancement.

TECHNICAL PRESENTATION CHECKLIST

TALENTSPHERE ELEVATE – TECHNICAL PRESENTATION CHECKLIST

1. PROJECT OVERVIEW

Project Name: TalentSphere Elevate

Project Type: AI-Powered Career Growth & Recruitment Platform

Users: High School Students, College Students, Working Professionals

Goal: Career discovery, placement support, and professional growth

2. TECHNOLOGY STACK

Layer	Technology
Frontend	Streamlit / React
Backend	FastAPI
Database	PostgreSQL / SQLite
AI / NLP	spaCy, NLTK, scikit-learn
Resume Parsing	pdfplumber, PyPDF2, python-docx
Visualization	Plotly
Authentication	JWT
Deployment	Render / Railway / AWS

3. USER MODULES

Module	Main Features
High School	Career discovery, aptitude analysis, learning roadmap

College	Resume builder, coding tests, placement score
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Professional	Upskilling, promotion analysis, salary insights
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4. DATABASE TABLES

- **users** – login & roles
- **high_school_profiles** – school student data
- **college_profiles** – college student data
- **professional_profiles** – professional data
- **skills** – master skill list
- **user_skills** – skill mapping
- **jobs** – job postings
- **applications** – job applications
- **assessments** – test results
- **career_recommendations** – AI outputs

5. AI COMPONENTS (IMPORTANT FOR VIVA)

AI Feature	Purpose
Resume Parsing	Extract skills, education, experience
Skill Extraction	Identify keywords from resume
Job Matching	Compare user skills with job requirements
Career Recommendation	Suggest suitable career paths
Skill Gap Analysis	Find missing skills
Placement Readiness	Calculate industry readiness score
Promotion Readiness	Evaluate professional growth potential

6. AI MATCHING FORMULA

Skills Match

40%

Experience Match

30%

Education Match

20%

Certifications

10%

Final Match Score

0–100

7. PROJECT ARCHITECTURE

Frontend

Streamlit / React

Backend API

FastAPI

AI Engine

NLP + Matching Logic

Database

PostgreSQL

8. API ENDPOINTS TO MENTION

Endpoint	Purpose
POST /register	create account

POST /login	authenticate user
POST /upload-resume	upload resume
POST /analyze-resume	AI analysis
GET /recommend-career	career suggestions
GET /job-matches	job matching
GET /dashboard	analytics data

9. SECURITY FEATURES

- **JWT Authentication**
- **Password Hashing**
- **Role-Based Access Control**
- **File Validation for Resume Upload**
- **Input Sanitization**
- **Protected API Routes**

10. PERFORMANCE TARGETS

Metric	Target
Concurrent Users	100+
Resume Processing Time	< 5 seconds
Job Match Generation	< 2 seconds
Applications Supported	1,000+
Dashboard Load Time	< 3 seconds

11. DEMO FLOW (MOST IMPORTANT)

1

Login as High School Student

2

Complete career assessment

3

Show AI career recommendation

4

Login as College Student

5

Upload resume

6

Display ATS score and skill gap

7

Show placement readiness score

8

Login as Working Professional

9

Display promotion readiness and salary growth

10

Show final AI recommendations

12. QUESTIONS REVIEWERS MAY ASK

Question

Short Answer

Why FastAPI?	High performance, async support, easy API development
Why PostgreSQL?	Reliable relational database with strong query support
How is AI used?	NLP for skill extraction and recommendation generation
How is job matching calculated?	Weighted score using skills, experience, education, certifications
What makes this project unique?	Supports school students, college students, and professionals in one platform

13. FINAL SLIDE CHECKLIST

Problem Statement

Objectives

Technology Stack

Architecture Diagram

Database Schema

AI Workflow

User Modules

API Endpoints

Security Features

Results & Screenshots

Future Enhancements

Conclusion

BEST ONE-LINE TECHNICAL SUMMARY FOR PRESENTATION

“TalentSphere Elevate is a full-stack AI-powered career intelligence platform built using FastAPI, PostgreSQL, and NLP technologies that provides personalized career recommendations, placement readiness analysis, resume screening, skill gap detection, and professional growth insights for school students, college students, and working professionals.”

WORKING MODEL EXPLANATION

Below is the WORKING MODEL EXPLANATION in simple TXT format that you can directly use in your project report, PPT, viva, or internship presentation.

TALENTSPHERE ELEVATE – WORKING MODEL EXPLANATION

PROJECT: TalentSphere Elevate – AI-Powered Career Growth & Recruitment Platform

1. OVERALL WORKING OF THE PROJECT

TalentSphere Elevate works as a multi-user AI platform that provides personalized career guidance, placement support, and professional growth recommendations. The system supports three different user types:

- **High School Students**
- **College Students**
- **Working Professionals**

Each user receives a customized dashboard and AI-generated recommendations based on their profile, skills, interests, and goals.

2. SYSTEM WORKFLOW

Step-by-Step Flow

End-to-End

1

User Registration

The user creates an account and selects a role: High School Student, College Student, or Working Professional.

2

Profile Creation

The user enters personal details, education, skills, interests, and career goals.

3

AI Analysis

The AI engine analyzes the profile, assessments, and uploaded resume (if available).

4

Recommendation Generation

The system generates career suggestions, skill gap analysis, learning paths, and job matches.

5

Dashboard Display

The user sees personalized results, scores, progress tracking, and recommended actions.

6

Continuous Improvement

As the user updates skills, completes assessments, or uploads a new resume, the AI recalculates recommendations.

3. HIGH SCHOOL STUDENT WORKING MODEL

Input:

- Class (9–12)
- Favorite subjects
- Hobbies
- Career interests
- Aptitude test answers

AI Processing:

- Interest analysis
- Personality evaluation
- Aptitude scoring
- Future career prediction

Output:

- Recommended careers

- **Future skills roadmap**
- **Learning recommendations**
- **Goal tracking dashboard**

Example:

Interest Score

Technology: 92%

Aptitude Score

Logical Thinking: 95%

AI Recommendation

AI Engineer

Next Skill

Learn Python

4. COLLEGE STUDENT WORKING MODEL

Input:

- **Degree and department**
- **Skills**
- **CGPA**
- **Projects**
- **Uploaded resume**
- **Coding test results**

AI Processing:

- **Resume parsing**
- **Skill extraction**
- **ATS score calculation**
- **Placement readiness analysis**
- **Job matching**

Output:

- **ATS resume score**

- **Missing skills**
- **Placement readiness score**
- **Internship recommendations**
- **Job matches**

Example:

Resume ATS Score

85%

Placement Score

78/100

Missing Skills

SQL, Data Structures

Top Job Match

Software Developer (88%)

5. WORKING PROFESSIONAL MODEL

Input:

- **Current role**
- **Experience**
- **Technical skills**
- **Leadership experience**
- **Career goals**
- **Certifications**

AI Processing:

- **Professional skill assessment**
- **Career transition analysis**
- **Promotion readiness calculation**
- **Salary benchmark comparison**
- **Industry trend analysis**

Output:

- Next best role
- Promotion readiness score
- Salary growth estimate
- Certification recommendations
- Advanced job matches

Example:

Current Role

Software Developer

Next Role

Senior Backend Engineer

Promotion Readiness

78%

Salary Growth

+25%

6. TECHNICAL WORKING ARCHITECTURE

Frontend

Streamlit / React Dashboard

Sends user data

Backend API

FastAPI

Processes requests

AI Engine

NLP + Recommendation Logic

Stores & retrieves

Database

PostgreSQL

7. HOW THE AI ENGINE WORKS

Step 1 – Data Collection

Collect profile details, assessments, and resume data.

Step 2 – Feature Extraction

Extract skills, education, interests, and experience.

Step 3 – Scoring

Skills Match

40%

Experience Match

30%

Education Match

20%

Certifications

10%

Final Match Score

0–100

Step 4 – Recommendation

Generate career, learning, internship, or job recommendations.

Step 5 – Dashboard Update

Display personalized results to the user.

8. END-TO-END EXAMPLE

1

College student uploads resume.

2

FastAPI receives the file.

3

Resume parser extracts skills.

4

AI compares skills with job requirements.

5

System calculates ATS score and placement score.

6

Missing skills are identified.

7

Internship and job recommendations are generated.

8

Dashboard shows final results.

9. OUTPUT SHOWN TO USER

User Type	AI Output
High School	Career recommendation + learning roadmap
College	ATS score + placement score + job matches
Professional	Promotion readiness + salary growth + next role

10. FINAL WORKING MODEL SUMMARY

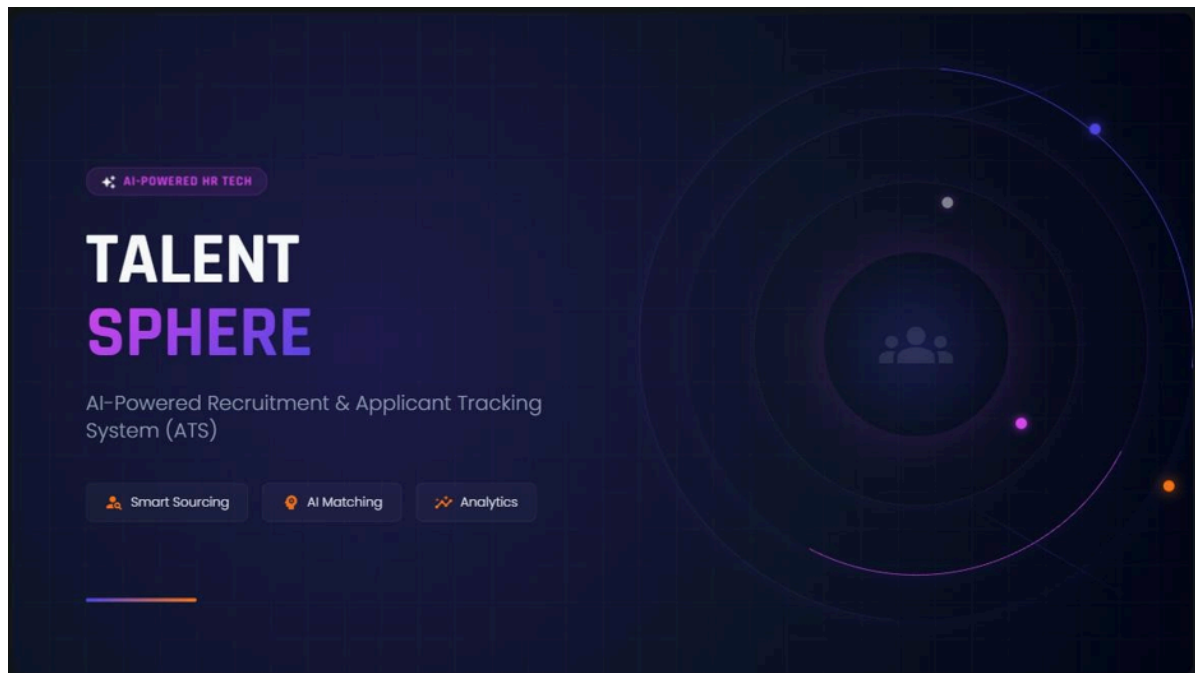
TalentSphere Elevate works by collecting user profile data, analyzing it using AI and NLP techniques, extracting relevant skills and career information, calculating readiness and matching scores, identifying skill gaps, and finally generating personalized career, placement, internship, and professional growth recommendations through role-based dashboards for High School Students, College Students, and Working Professionals.

BEST 2-LINE EXPLANATION FOR VIVA

“The system accepts user data and resumes, processes them through a FastAPI backend and AI recommendation engine, and generates personalized career guidance, placement analysis, and professional growth insights. The entire workflow is role-based, allowing school students, college students, and working professionals to receive different AI-driven recommendations from the same platform.”

EXAMPLE FILE

Talent Sphere – AI-Powered Recruitment & Applicant Tracking System (ATS)



2. Project Statement

Recruitment is one of the most critical business processes for organizations, yet many companies still rely on manual resume screening, fragmented communication channels, and inefficient hiring workflows. This often results in delayed hiring decisions, poor candidate experience, and increased recruitment costs.

TalentSphere is an AI-powered Recruitment and Applicant Tracking System (ATS) designed to streamline the entire hiring lifecycle. The platform enables recruiters to create job postings, manage applications, track candidate progress, schedule interviews, and make data-driven hiring decisions. AI-driven resume screening automatically evaluates candidates against job requirements, ranks applicants, identifies skill gaps, and recommends the most suitable candidates.

The system provides role-based access for recruiters, hiring managers, and candidates through an intuitive web interface. With analytics dashboards, automated notifications, and intelligent candidate matching, TalentSphere improves hiring efficiency, reduces recruitment time, and enhances workforce acquisition strategies.

3. Outcomes

- AI-powered resume screening and candidate ranking system.
- Centralized Applicant Tracking System (ATS) for managing recruitment workflows.
- Intelligent job-candidate matching based on skills, experience, and qualifications.
- Automated interview scheduling and email notifications.
- Recruitment analytics dashboard with hiring KPIs and funnel metrics.
- Candidate portal for job applications and status tracking.
- Recruiter dashboard for managing jobs, applications, and interview processes.
- AI-based skill gap analysis and hiring recommendations.
- Role-based authentication and secure recruitment management.
- Exportable recruitment reports and hiring insights.

4. Modules to be Implemented

1. User Management & Authentication

- Secure registration and login for Recruiters, Hiring Managers, and Candidates.
- Role-based access control.
- Profile management and account settings.
- Resume and document uploads.

2. Job Management System

- Create, update, and publish job postings.
- Define required skills, qualifications, and experience levels.
- Manage active, closed, and draft job openings.
- Job search and filtering functionalities.

3. Applicant Tracking System (ATS)

- Track candidates across recruitment stages.
- Application status management.
- Recruiter notes and candidate evaluation.
- Candidate pipeline visualization.

4. AI Resume Screening Engine

- Resume parsing using NLP techniques.
- Skill extraction and keyword identification.
- Resume-job matching score calculation.
- Automated candidate ranking and recommendations.

5. Interview Management

- Interview scheduling and calendar integration.
- Automated interview invitations.
- Interview feedback collection.
- Multi-stage interview tracking.

6. Recruitment Analytics Dashboard

- Hiring funnel analysis.
- Application source tracking.
- Time-to-hire analysis.
- Candidate conversion metrics.
- Recruiter performance reports.

7. Notification & Communication Module

- Automated email notifications.
- Application status updates.
- Interview reminders.
- Recruiter-candidate communication tracking.

8. AI Insights & Recommendation Engine

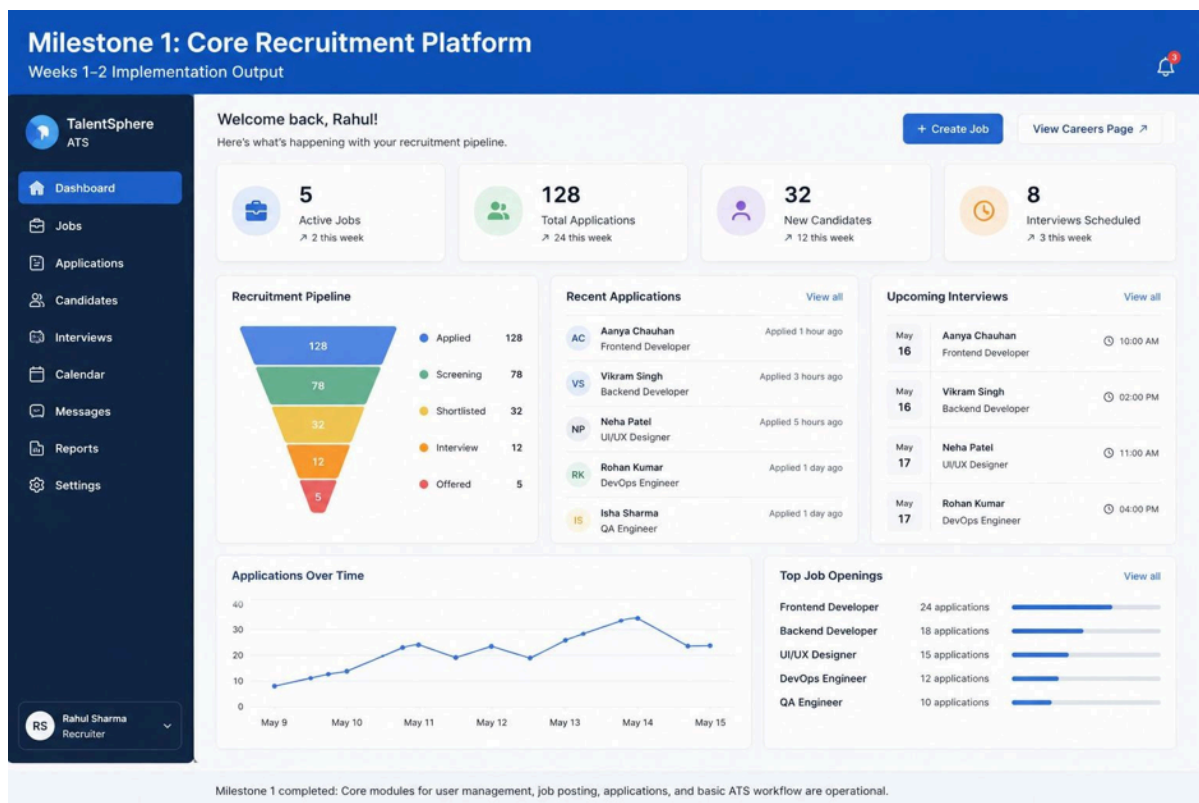
- Skill gap analysis.
- Candidate suitability predictions.
- Hiring trend analysis.
- Recruitment forecasting insights.

5. Week-wise Module Implementation &

High-Level Requirements Milestone 1

(Weeks 1–2) – Core Recruitment Platform

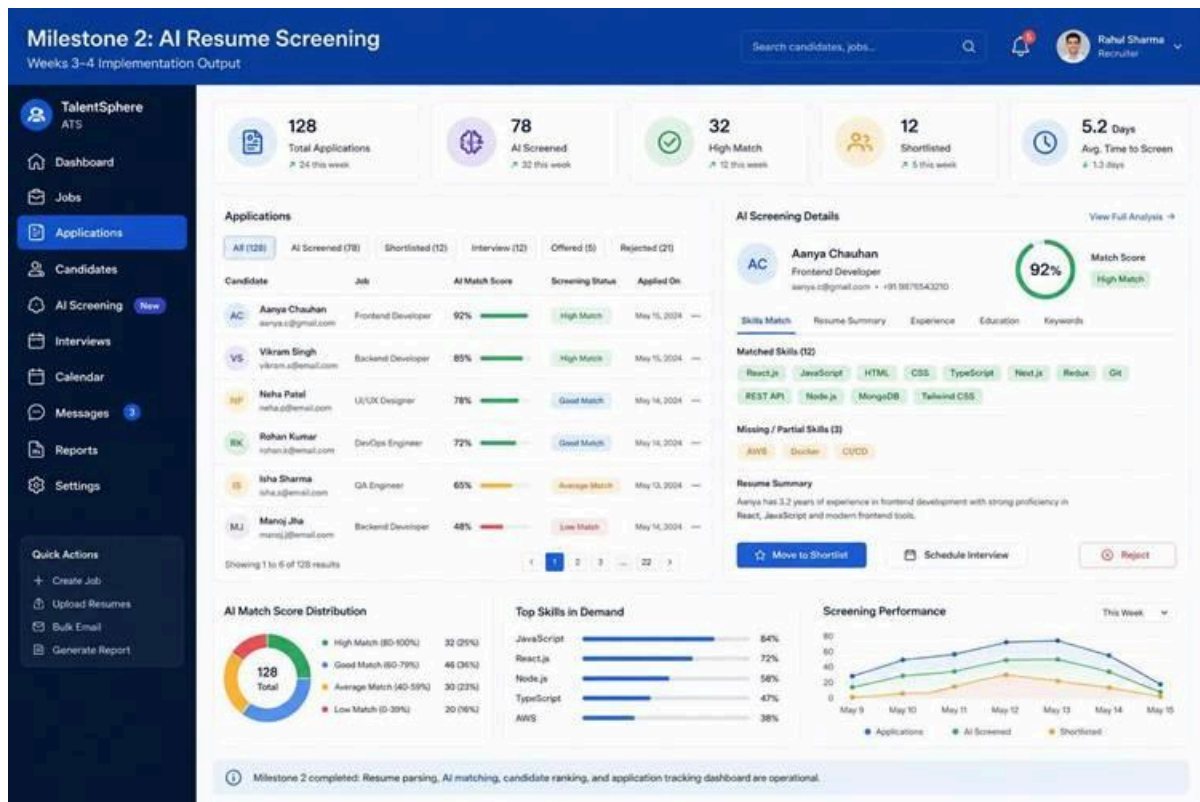
- Develop user authentication and role management.
- Build candidate registration and profile creation.
- Implement job posting and job application modules.
- Create basic ATS workflow for application tracking.
- Design initial recruiter and candidate dashboards.



Milestone 2 (Weeks 3-4) – AI Resume Screening

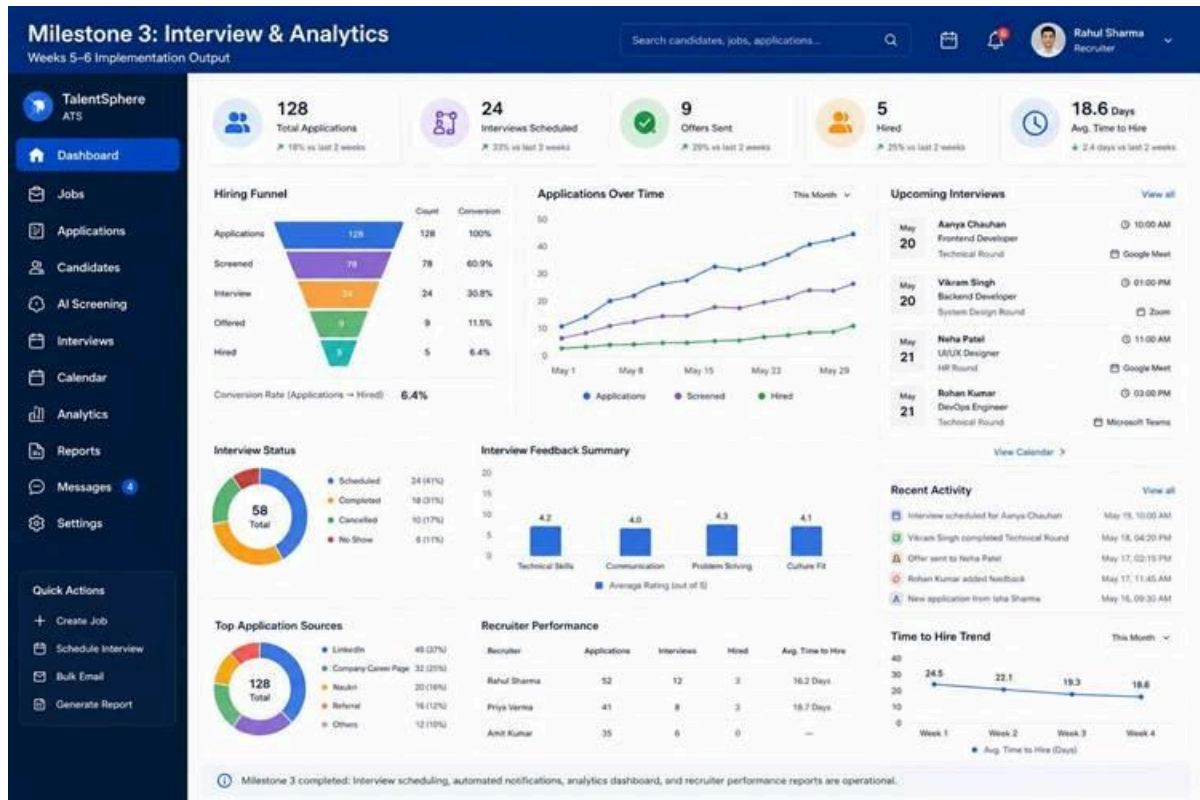
- Implement resume upload and parsing functionality.
- Extract skills, qualifications, and experience using NLP.
- Develop candidate-job matching algorithm.

- Generate AI-based candidate ranking scores.
- Build application tracking dashboard.



Milestone 3 (Weeks 5-6) – Interview & Analytics

- Develop interview scheduling system.
- Implement email notification services.
- Build recruitment analytics dashboard.
- Add hiring funnel and candidate conversion reports.
- Create recruiter performance metrics.



Milestone 4 (Weeks 7–8) – AI Insights & Final Deployment

- Implement AI recommendation engine.
- Add skill gap analysis module.
- Generate hiring intelligence reports.
- Optimize performance and security.
- Deploy application and prepare documentation.



6. Evaluation Criteria

Milestone 1 (Week 2)

- User authentication system fully operational.
- Job posting and application workflows functional.
- ATS pipeline successfully tracks candidate progress.
- Candidate and recruiter dashboards accessible.

Milestone 2 (Week 4)

- Resume parser extracts candidate information accurately.
- AI matching engine generates candidate scores.
- Candidate ranking system operational.

- ATS dashboard provides real-time application status.

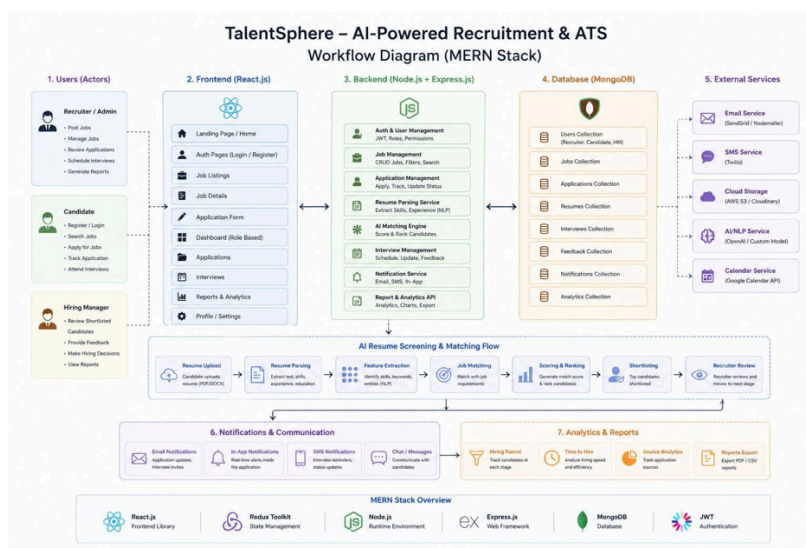
Milestone 3 (Week 6)

- Interview scheduling system functional.
- Automated notifications delivered successfully.
- Recruitment analytics dashboard operational.
- Hiring funnel and recruiter performance reports available.

Milestone 4 (Week 8)

- AI recommendation engine provides hiring insights.
- Skill gap analysis generates meaningful recommendations.
- End-to-end recruitment workflow operational.
- System supports 1,000+ applications efficiently.
- Documentation and deployment completed.

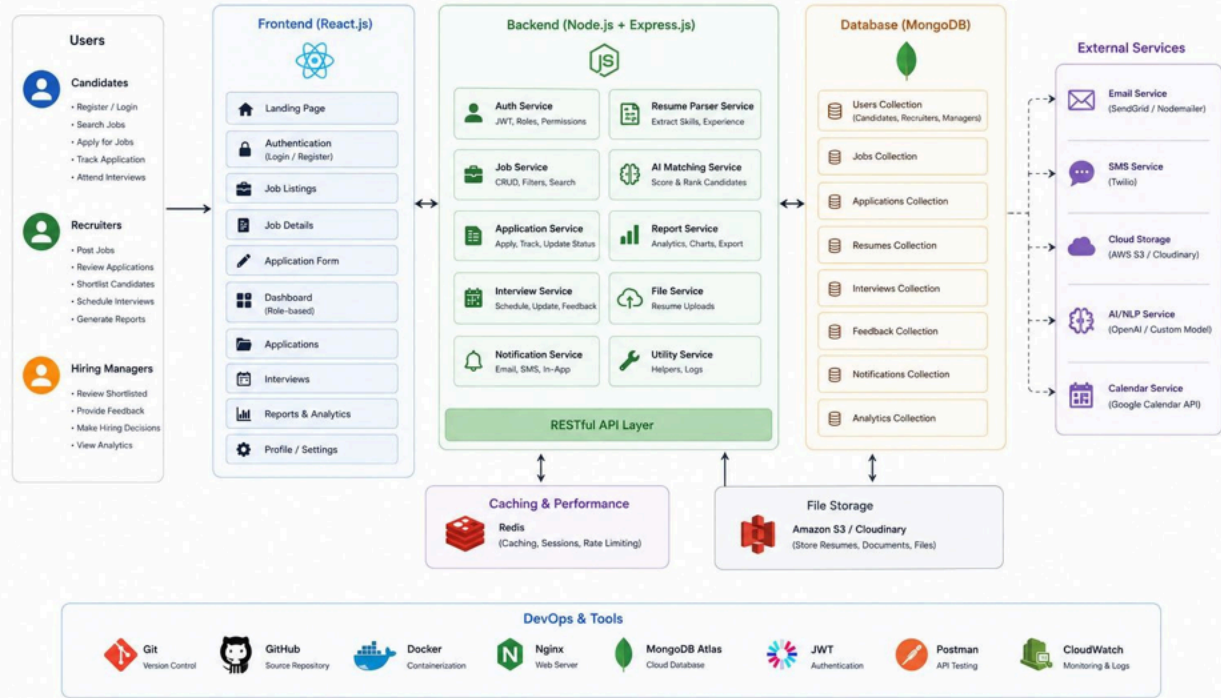
7. Workflow Diagram



8. Architecture Diagram

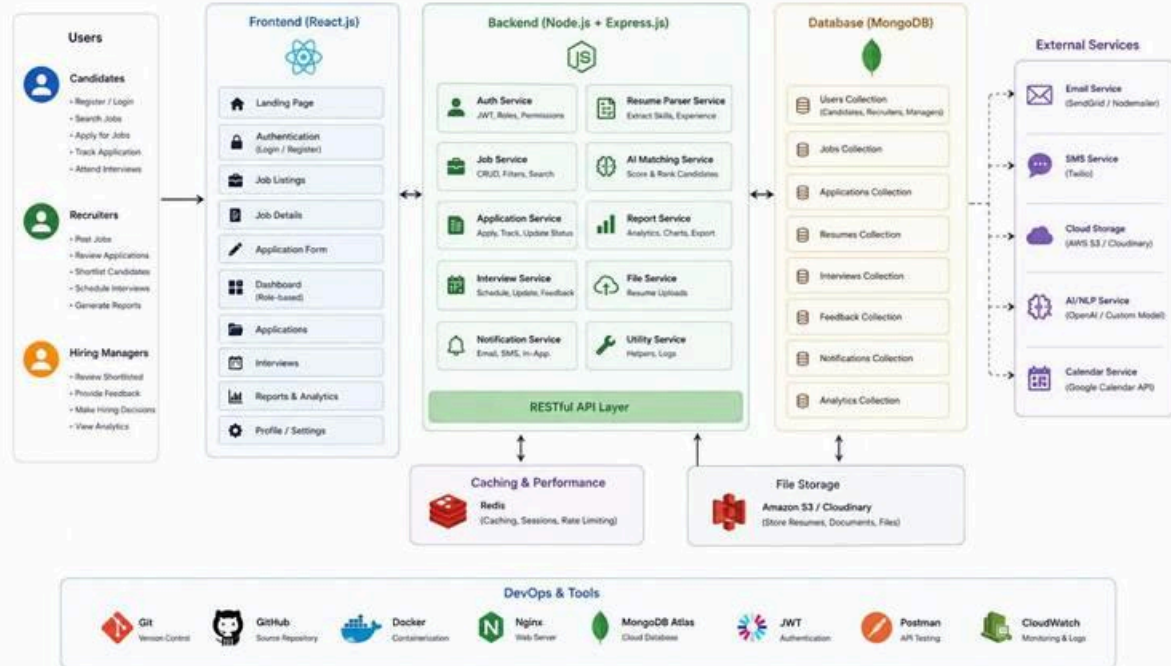
TalentSphere – AI-Powered Recruitment & ATS

Architecture Diagram (MERN Stack)

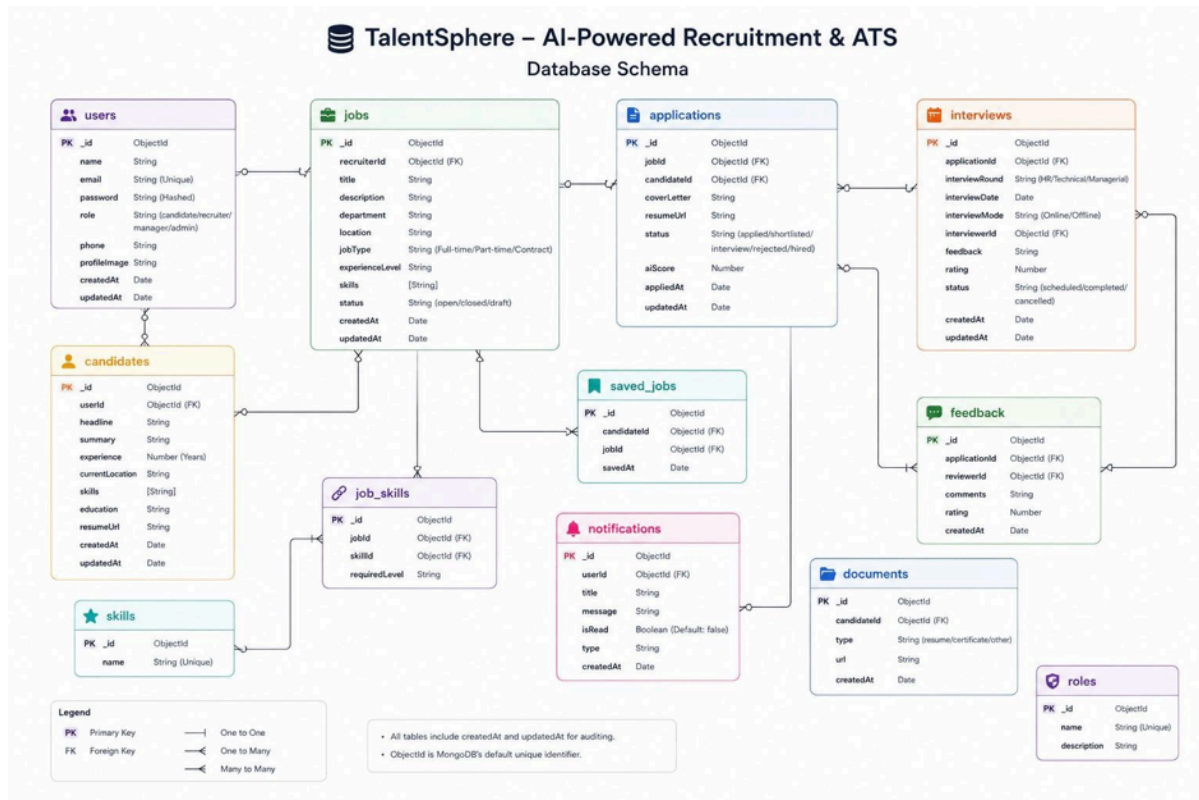


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Architecture Diagram (MERN Stack)



9. Database Schema



Tab 8

